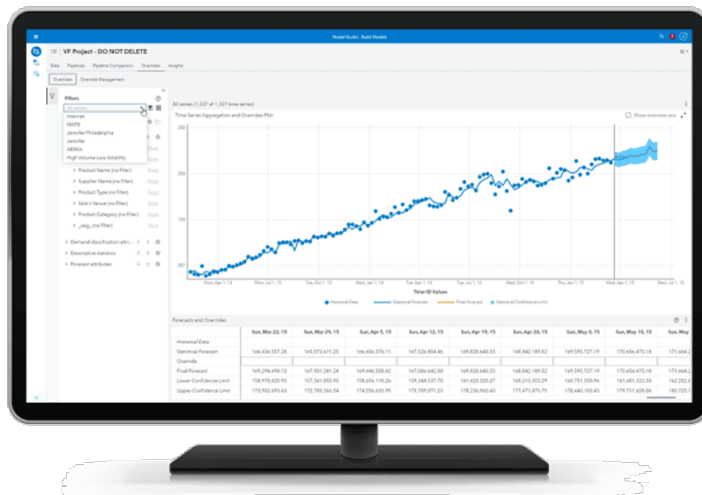


TECNOLÓGICO NACIONAL DE MÉXICO

INSTITUTO TECNOLÓGICO DE CIUDAD GUZMÁN

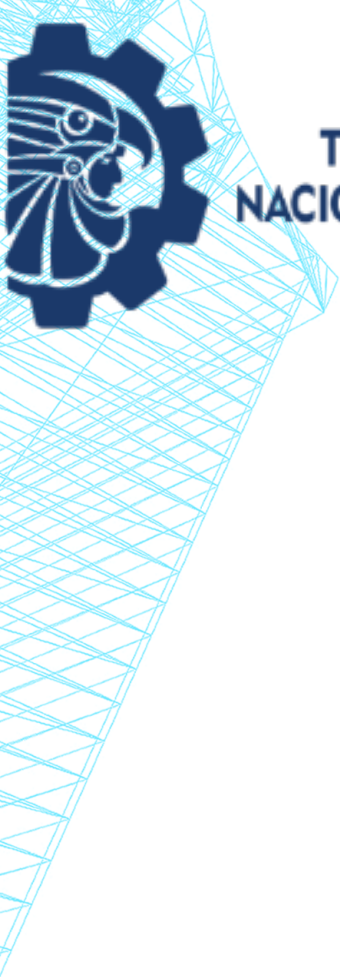
Temas Selectos III

Maestría en Ciencias Computacionales 3er Semestre



Presenta
Dr. Julio Alberto García Rodríguez

Ciudad Guzmán, Agosto de 2026



TECNOLÓGICO
NACIONAL DE MÉXICO

SITIO OFICIAL

<https://cdguzman.tecnm.mx/pag/>

EDUCACIÓN

SECRETARÍA DE EDUCACIÓN PÚBLICA



The screenshot shows the official website of TecNM Ciudad Guzmán. The header includes the logos of the Mexican Government, the Ministry of Education, and TecNM. Navigation links for 'Trámites', 'Gobierno', and a search icon are present. A secondary navigation bar lists various services: 'Conoce', 'Nuevo Ingreso', 'Oferta Educativa', 'Eventos y Reconocimientos', 'Galería', 'ITCG', 'Recursos Humanos', and 'Reglamentos'. The main banner features the text 'ESTUDIA UNA MAESTRÍA' in large white letters on a dark blue background, with 'MAESTRÍA' highlighted in a red speech bubble. Below this, a red box lists the master's programs: 'Maestría en: Ingeniería Electrónica (MIE)' and 'Ciencias de la Computación (MCC)'. To the right, a dark blue box says 'En el TecNM Ciudad Guzmán'. At the bottom, a red button with a hand icon and the text 'Clic aquí' is positioned next to the text 'Requisitos y más información'.

Gobierno de México

Trámites Gobierno

Estudiantes Egresados Avisos Estadística Transparencia Protección de Datos Personales

Conoce Nuevo Ingreso Oferta Educativa Eventos y Reconocimientos Galería ITCG Recursos Humanos Reglamentos

ESTUDIA UNA MAESTRÍA

Maestría en:

- Ingeniería Electrónica (MIE)
- Ciencias de la Computación (MCC)

En el TecNM Ciudad Guzmán

Requisitos y más información

Clic aquí

ENCUADRE DE LA MATERIA

- **Actividades de aprendizaje** **30%**
(Tareas, Ejercicios de clase, Investigaciones, Prácticas, Exposiciones)
- **Instrumento de Evaluación: Examen escrito** **40%**
- **Proyecto de Asignatura** **30%**

A través del presente escrito CONFIRMO que conozco el ENCUADRE y el PROGRAMA DE ESTUDIOS de la materia de TÓPICOS SELECTOS

Nombre completo y firma

Fecha

DATOS DE CONTACTO

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Oficina: Edificio Ñ, Cubiculo Jefatura de Ciencias básicas.

341-110-8223

CLASS BEST PRACTICES

- Limited Smartphone use. Emergencies only.
- Put attention in class, avoid distractors as Social Networks.
- Ask when you feel lost.
- You should meet the prerequisites. (Math, Programming, Soft skills).
- All resources must be in English.
- Some classes will be conducted entirely in English by both the instructor and the students.

Class Best practices

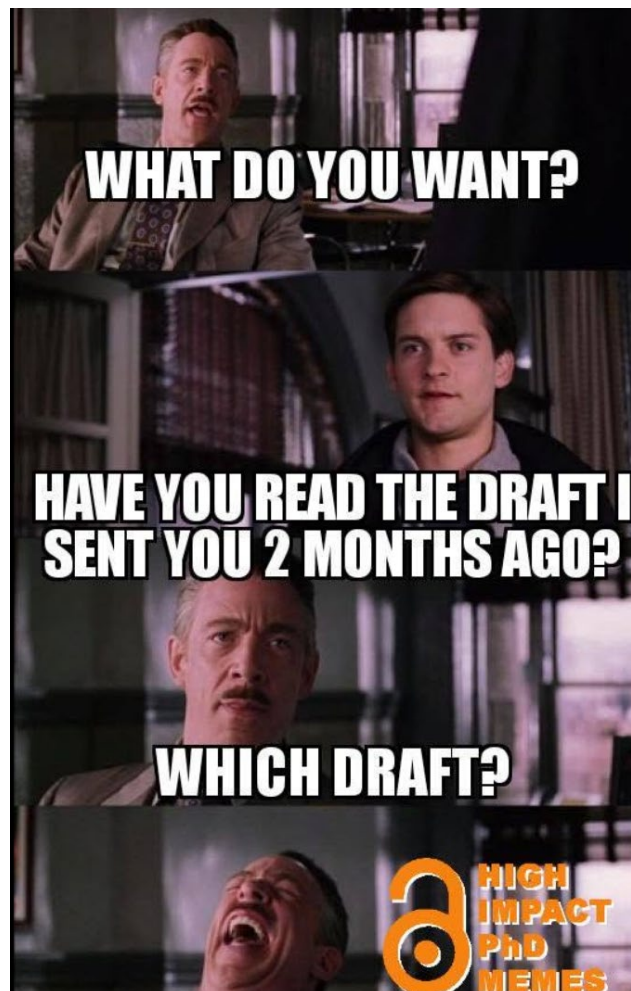
- Remember that Natural Intelligence is still better than Artificial Intelligence. Use AI tools wisely, responsibly, and ethically.
- The class focuses on developing logical, mathematical, and computational thinking skills. It is not focused on developing thousands of lines of code that even the student doesn't understand.
- Get your hands dirty.
- Stay engaged.

Class Best practices

Me sacrificing my sleep, sanity and soul for
my research,

Supervisor: This is not enough, you need to
work harder 😊

Me-



1. Introduction

1.1 Time Series Fundamentals

Forecasting

Forecasting has fascinated people for thousands of years, sometimes being considered a sign of divine inspiration, and sometimes being seen as a criminal activity. The Jewish prophet Isaiah wrote in about 700 BC.

Isaiah 41:23 ▾ New International Version ▾

23 tell us what the future holds,
 so we may know that you are gods.
Do something, whether good or bad,
 so that we will be dismayed and filled with fear.

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One hundred years later, in ancient Babylon, forecasters would foretell the future based on the appearance of a sheep's liver.



Babylonian clay tablet of a sheep's liver used for divination (collection British Museum, London).

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9944526/#r1>

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People wanting forecasts would journey to Delphi in Greece to consult the Oracle, who would provide her predictions while intoxicated by ethylene vapours.



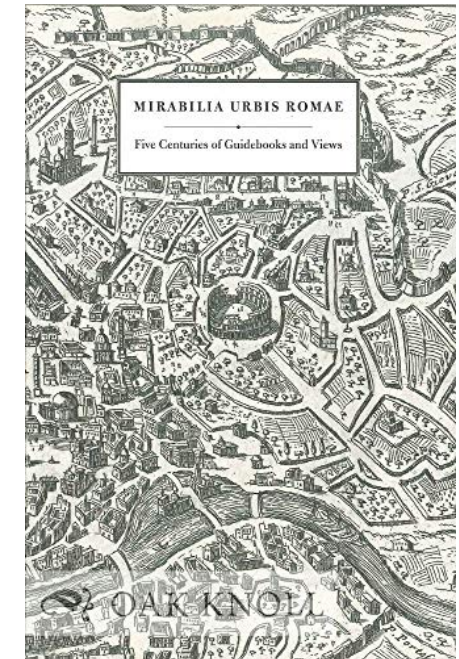
Pythia was the title of the high priestess of the Temple of Apollo at Delphi in central Greece. She served as its oracle and was known as the Oracle of Delphi. Her title was sometimes historically glossed in English as the Pythoness.

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Octavian Augustus (63 b.C. – 14 a.C.) had become extremely powerful during the Pax Romana. Because of his success, the Roman Senate wanted to honor him as “divus”, giving him divine honors. Augustus initially resisted this idea.

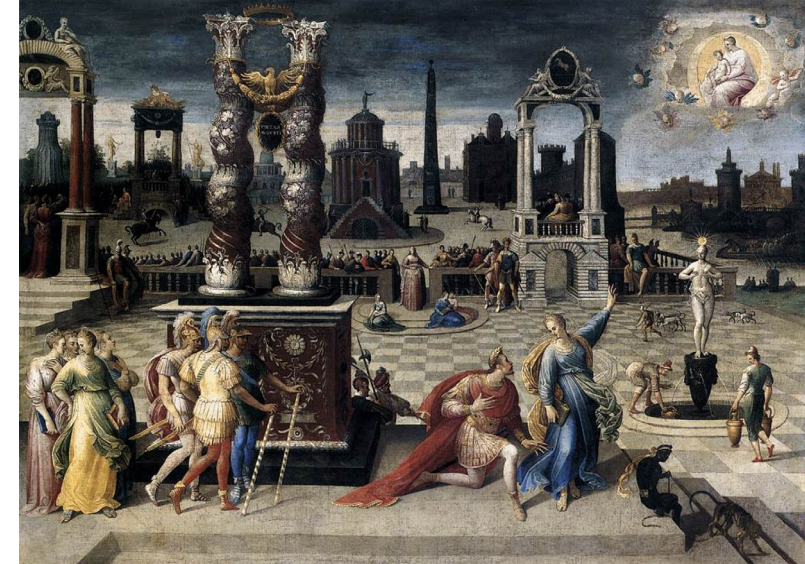


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To seek guidance, Augustus went to Tibur (modern Tivoli) and consulted the Tiburtine Sibyl, a pagan prophetess. After three days of fasting, the Sibyl announced that a heavenly King would come into the world. Augustus then received a vision of a beautiful Virgin with a Child in her arms, standing above an altar. A heavenly voice identified the vision as connected with the Son of God.

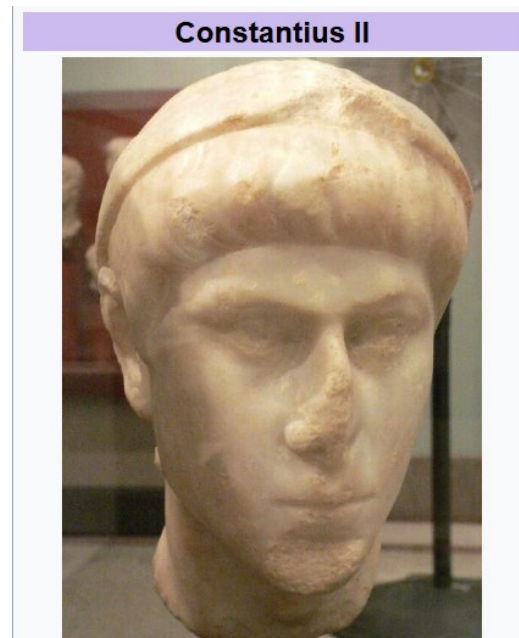


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Forecasters had a tougher time under the emperor Constantius II, who issued a decree in 357 A.D forbidding anyone “to consult a soothsayer, a mathematician, or a forecaster ... May curiosity to foretell the future be silenced forever.”



Possible head portrait of Constantius II found in modern [al-Bab, Syria](#) ([Penn Museum](#))^{[1][2][3]}

Roman emperor

Augustus 9 September 337 –
3 November 361

Predecessor [Constantine I](#)

Successor [Julian](#)

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5. The Emperor Constantius and the Caesar Julian to the People.

No one shall consult an haruspex, a diviner, or a soothsayer, and wicked confessions made to augurs and prophets must cease. Chaldeans, magicians, and others who are commonly called malefactors on account of the enormity of their crimes, shall no longer practice their infamous arts. Let all curiosity relating to divination be at an end, for anyone who hereafter refuses obedience to Our orders shall be put to death, and laid low with the avenging sword.

Given at Milan, on the eighth of the *Kalends* of February, during the Consulate of Constantius, Consul for the eighth time, and the Caesar Julian, 357.

https://droitromain.univ-grenoble-alpes.fr/Anglica/CJ9_Scott.gr.htm#9

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CTh.9.16.4 [=brev.9.13.2]

Imp. constantius a. et iulianus c. ad populum. nemo haruspicem consulat aut mathematicum, nemo hariolum. augurum et vatum prava confessio conticescat. chaldaei ac magi et ceteri, quos maleficos ob facinorum magnitudinem vulgus appellat, nec ad hanc partem aliquid moliantur. sileat omnibus perpetuo divinandi curiositas. etenim supplicium capitis feret gladio ultore prostratus, quicumque* iussis obsequium denegaverit. dat. viii. kal. febr. mediolano, constantio a. ix. et iuliano caes. ii. coss.

<https://www.thelatinlibrary.com/theodosius/theod09.shtml>

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Forecasting

A similar ban on forecasting occurred in England in 1824 when “every person pretending or professing to tell fortunes” was “deemed a rogue and vagabond”. The punishment was up to three months’ imprisonment with hard labour.

IV. [Persons committing certain Offences to be deemed Rogues and Vagabonds.] And be it further enacted, That every Person committing any of the Offences herein-before mentioned, after having been convicted as an idle and disorderly Person; every Person pretending or professing to tell Fortunes, or using any subtle Craft, Means, or Device, by Palmistry or otherwise, to deceive and impose on any of His Majesty's Subjects; every Person wandering abroad and lodging in any Barn or Outhouse, or in any deserted or unoccupied Building, or in the open Air, or under a Tent, or in any Cart or Waggon, not having any visible Means of Subsistence, and not giving a good Account of himself or herself; every Person wilfully exposing to view, in any Street, Road, Highway, or Public Place, any obscene Print, Picture, or other indecent Exhibition: every

<https://statutes.org.uk/site/the-statutes/nineteenth-century/5-geo-iv-c-83-vagrancy-act-1824/>

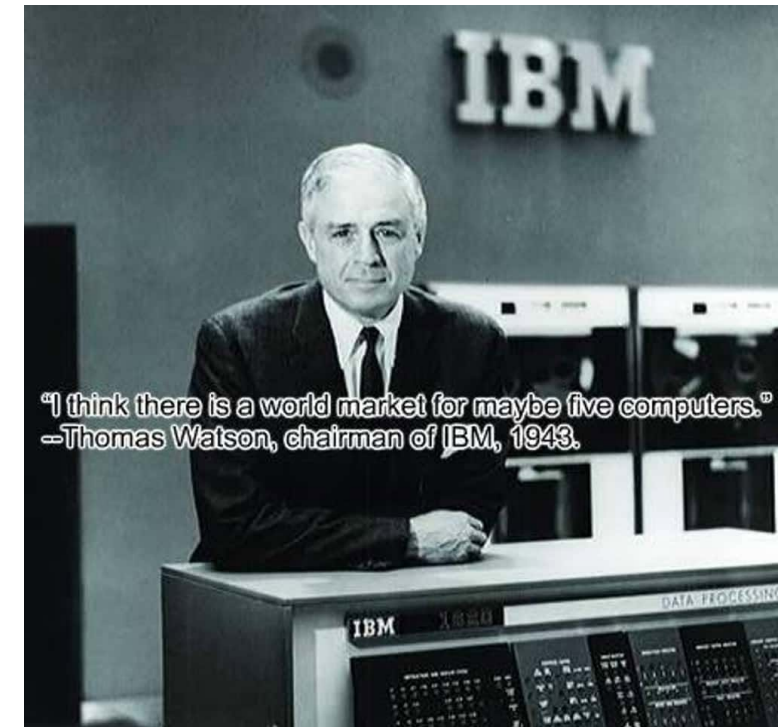
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The varying fortunes of forecasters arise because good forecasts can seem almost magical, while bad forecasts may be dangerous. Consider the following famous predictions about computing.

- I think there is a world market for maybe five computers. (Chairman of IBM, 1943)



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- Computers in the future may weigh no more than 1.5 tons. (Popular Mechanics, 1949)

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Forecasting

- There is no reason anyone would want a computer in their home. (President, DEC, 1977)



“There is no reason anyone would want a computer in their home.” -

Ken Olson, president, chairman and founder of Digital Equipment Corp. (DEC), maker of big business mainframe computers, arguing against the PC in 1977

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This quote was made only three years before IBM produced the first personal computer. Not surprisingly, you can no longer buy a DEC computer.

Forecasting is obviously a difficult activity, and businesses that do it well have a big advantage over those whose forecasts fail.



“There is no reason anyone would want a computer in their home.” -

Ken Olson, president, chairman and founder of Digital Equipment Corp. (DEC), maker of big business mainframe computers, arguing against the PC in 1977